5 Remove left Recursive by substitution

$A \to A\alpha \mid \beta$

add $B \to \alpha \mid \alpha B$ 左递归→右递归

then $A \to \beta B \mid \beta$

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82 Pumping Lemma for CFG

pumping (for CFL) is used to prove that a Language is NOT Context Free

: If A is a Context Free Language, then, A has a Pumping Length "P" such that any string 'S', where $|S| \geq P$ may be divided into 5 pieces $S = uvxyz$ such that:

(1) uv^ixy^iz is in A for every $i \geq 0$

(2) $|vy| > 0$

(3) $|vxy| \leq p$

Steps:

- → Assume that A is Context Free
- → It has to have a Pumping Length P
- → All strings longor than P can be Pumped $|s| \geq P$
- → Now find a string 's' in A sunch that $|s| \geq P$
- → Divide S into uvxyz
- → show that $uv^ixy^iz \notin A$ for some i
- → Then consider the ways that S can be divided into uvxyz
- → show that none of these can satisfy all the 3 pumping conditions at the same time
- → S cannot be pumped == CONTRADICTION

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(83) Example of Pumping lemma of CFG

$L = \{ a^N b^N c^N \mid N \geq 0 \}$

Proof:

1. Assume L is context-free

2. Pumping Lemma: so There is a pumping Length $P > 0$ for L

3. we choose string $a^P b^P c^P$. since $|S| = 3p \geq p$, the Pumping Lemma says S can be divided into ~~ux^iy~~ uv^ixy^iz such that:

$$\begin{cases} |vxy| \leq p \\ |vy| > 0 \\ uv^ixy^iz \in L \text{ for all } i \geq 0 \end{cases}$$

4. Consider Cases:

case 1: vxy is in a^P ⇒ pumping by $i=0$, ~~so~~ from $|vy| > 0$, we know $i > 0$, so $S = a^{P-2i} b^P c^P$ and $P-2i < P$, so $S \notin L$

case 2: vxy is in b^P, similar with case 1, $S \notin L$

case 3: vxy is in c^P, similar with case 1, $S \notin L$

case 4: vxy crosses a^P and b^P, ⇒ $S = a^{P-i} b^{P-i} c^P \notin L$

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case 5: vxy crosses b^P and c^P: similar with case 4 ~~&~~ $\notin L$

case 6: vxy crosses $a^P b^P c^P$, this impossible, because $|vxy| \le P$, and $|a^P b^P c^P| = 3P$

5. Contradiction: In all possible ~~cases~~ cases, pumping s resules in a string that is not in L. This contradicts the Pumping Lemma

6. Conclusion: Therefore, our initial assumption that L is context-free must be false.

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(84) Example 2.

show that $L=\{ww \mid w \in \{0,1\}^*\}$ is NOT Context Free

1. Assume L is a Context free language
2. ~~So from~~ we choose $0^P1^P0^P_1 1^P_1$ as S, from pumping Lemma, S can be divided into uv^ixy^iz such that

$$\begin{cases} uv^ixy^iz \text{ for } i \geq 0 \quad S \in L \\ |vxy| \leq P \\ |vy| > 0 \end{cases}$$

3. Consider cases:

1. vxy is in 0^P, then chose $i=0 \Rightarrow$
$S = 0^{P-2i}1^P0^P1^P$, from $|vy|>0$, we can know $i>0$
so $p-2i \neq P$, $\Rightarrow S \notin L$
2. cases: vxy in $1^P, 0^P_1, 1^P_1$, all similar with case 1
$\Rightarrow S \notin L$
3. vxy is crosses in $0^P1^P \Rightarrow$ pumping i, cases $0^{P-i}1^{P-i}$, 0^P1^{P-2i}, $0^{P-2i}1^P$ all make $S \notin L$, in $0^P_1 1^P_1$ case is similiar with this. 4. cross, $0^P1^P0^P$, $1^P0^P1^P$, $0^P1^P0^P1^P$